Tank 3: Try three ideas

Goal: I can sketch different solutions and choose one using a clear reason.

35 minutes | 7 steps | Engineering and Design | Playbook A meeting 10, grade 6 path; Playbook B meeting 11, grade 6 path

Ask | **Plan** | Build | Test | Improve

You are done when

Show three ideas and explain why one is the best first test.

You will use

- Paper
- Pencils
- Rulers
- Reusable criteria cards

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 2 min

- Read today's goal.
- Look at the example.

2 Step 2: Tell your partner — 1 min

- Tell your partner one useful detail.

3 Step 3: Watch and try — 5 min

- See three different ways to solve one small problem.
- Compare usefulness, simplicity, and available materials.

4 Step 4: Make it work — 17 min

- Sketch three different solutions.
- Label the part that does the work.
- Circle one idea you can test with classroom materials.

5 Step 5: Check one thing — 5 min

- Show the ideas to a partner.
- Name one strength and one concern for the chosen idea.
- Revise the sketch before building.

6 Step 6: Reset together — 3 min

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

7 Step 7: Show your result — 2 min

- Show three ideas and explain why one is the best first test.

What to turn in

Three sketches, one chosen idea and the reason.

No computer? Another way to do it

Describe three ideas aloud while a partner sketches.

Finished early? Next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Stay safe

Use teacher-approved materials and tools. Keep paths clear. Return tools before moving.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.